Example 9.64

Find the volume of the spherical cap of height h cut of from a sphere of radius r.

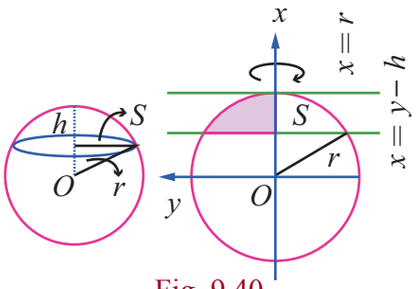


Fig. 9.40

Solution

If the region in the first quadrant bounded by the circle $x^2+y^2=r^2$, the x-axis, the lines $x=r-h$ and $x=r$ is revolved about the x-axis, then the solid generated is a spherical cap of height h cut of from a sphere of radius r. See Fig. 9.40. Hence, the required volume is given by

$$V = \pi\int_{r-h}^{r} y^2dx = \pi\int_{r-h}^{r}\left(r^2-x^2\right)dx = \pi\left(r^2x-\frac{x^3}{3}\right)_{r-h}^{r}$$

$$= \pi\left(r^2\left(r-\left(r-h\right)\right)-\frac{\left(r^3-\left(r-h\right)^3\right)}{3}\right) = \pi\left(r^2h-\frac{\left(r^3-\left(r^3-3r^2h+3rh^2-h^3\right)\right)}{3}\right)$$

$$= \pi\left(\frac{3rh^2-h^3}{3}\right) = \frac{1}{3}\pi h^2\left(3r-h\right).$$

Note

We can rewrite the above volume in terms of the radius of the cap.

If ρ is the radius of the cap, then $\rho^2+\left(r-h\right)^2=r^2$.

Then, we have $r=\dfrac{\rho^2+h^2}{2h}$. Eliminating r, we get

$$V=\frac{1}{3}\pi h^2\left[3\left(\frac{\rho^2+h^2}{2h}\right)-h\right]=\frac{1}{3}\pi h\left[\left(\frac{3\rho^2+h^2}{2}\right)\right]=\frac{1}{6}\pi h\left(3\rho^2+h^2\right).$$

Example 9.65

Find the volume of the solid formed by revolving the region bounded by the parabola $y=x^2$, x-axis, ordinates $x=0$ and $x=1$ about the x-axis.

Solution

The region to be revolved about the x-axis is sketched as in Fig.9.41. Hence, the required volume is given by

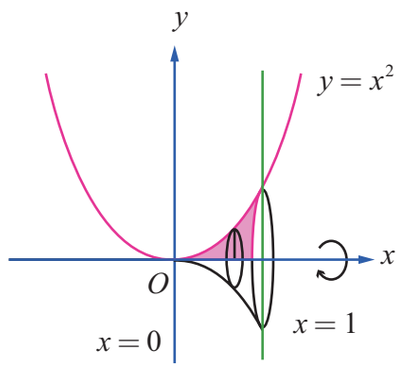


Fig. 9.41

$$V = \pi\int_0^1 y^2dx$$

$$= \pi\int_0^1\left(x^2\right)^2dx$$

$$= \pi\left[\frac{x^5}{5}\right]_0^1$$

$$= \pi\left[\frac{1}{5}-0\right]=\frac{\pi}{5}.$$

Example 9.66

Find the volume of the solid formed by revolving the region bounded by the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1, a>b$ about the major axis.

Chapter 9 Applications of Integration.indd 137 17-02-2020 17:08:24